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TAKE-HOME ASSIGNMENT · SUBMISSION

Procurement & Finance Automation

Raw-material reorder alerts and vendor-invoice matching — built as a live React console with Gemini extraction, a deterministic rules engine, and human approval on every money decision.

STACK	MATHS	TESTS	EVIDENCE
React · Vite · Node · Gemini	**100% in code**	**Python + JS, hand-checked**	**Live screenshots**

REQUIRED DELIVERABLE	SECTION
Part 1 — design answers	1
Issue 1 — build evidence, item results, alert, supplier email	2
Issue 2 — build evidence, exception report, two drafts, totals	3
Part 4 — reliability note (6–8 sentences)	4
Reproduction, validation and repository	5

Every screenshot in this document was captured from the running application using a live Gemini call — not mock-ups.

1 · Part 1 — Design

One principle runs through everything: **an LLM reads messy text; code does every number and status; humans approve anything that moves money.**

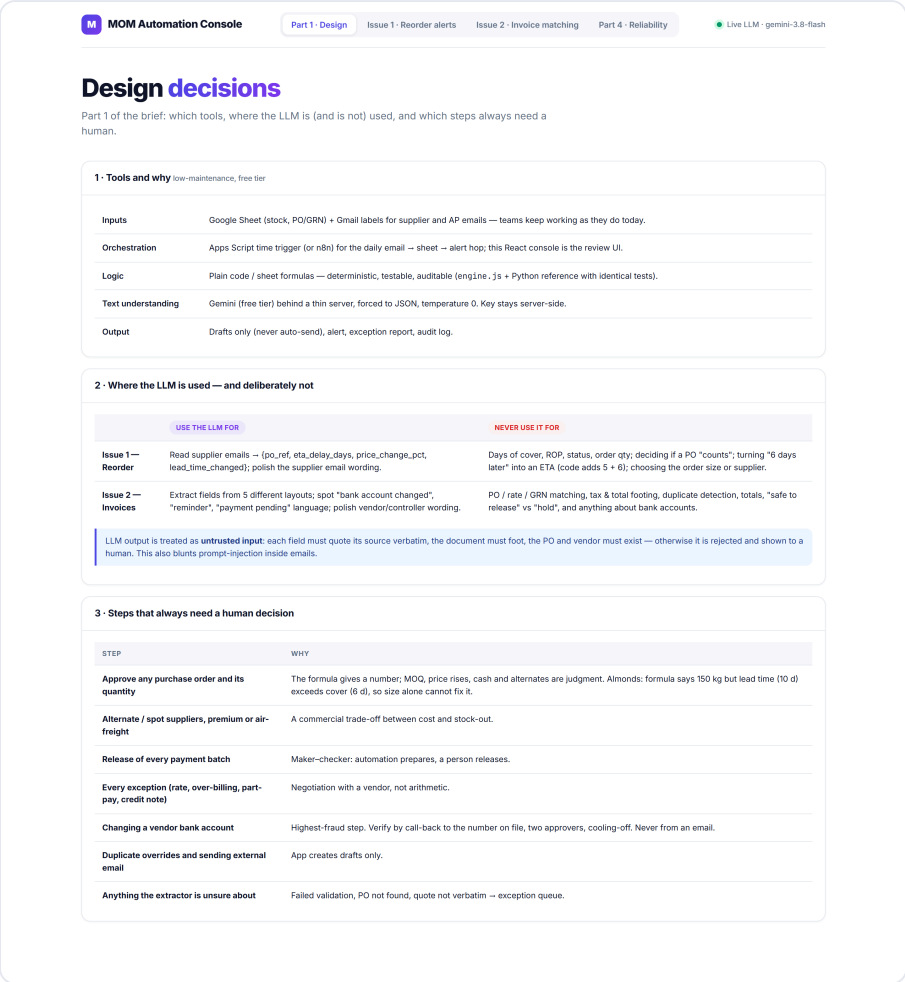


Figure 1 — Design tab of the console (Part 1 answers).

1.1 Which tools and why

LAYER	CHOICE AND REASON
Inputs	Google Sheet (stock, PO/GRN) plus Gmail labels for supplier and AP emails — teams keep working as they do today.
Orchestration	Apps Script time trigger or n8n for the daily email → sheet → alert hop; free, readable by non-engineers. The React console is the review UI.
Logic	Plain code (JS engine, mirrored by a Python reference). Deterministic, testable, auditable.
Text understanding	Gemini free tier behind a thin Node server, JSON-only output at temperature 0; key stays server-side.
Output	Drafts only (never auto-send), alert, exception report, audit log.

1.2 Where the LLM is used — and deliberately not

	USE THE LLM FOR	NEVER USE IT FOR
Issue 1	Reading supplier emails into `{po_ref, eta_delay_days, price_change_pct, lead_time_changed}` ; polishing the supplier email.	Days of cover, ROP, status, order qty; deciding whether a PO "counts"; converting "6 days later" into an ETA (code adds 5 + 6); choosing order size or supplier.
Issue 2	Extracting fields from 5 layouts; spotting "bank account changed / reminder / payment pending" language; polishing drafts.	PO/rate/GRN matching, footing of tax and totals, duplicate detection, totals, release vs hold, anything about bank accounts.

> LLM output is treated as **untrusted input**: quotes must appear verbatim in the source, documents must foot, POs and vendors must exist, and a second independent extractor must agree — otherwise it is rejected and shown to a human. This also blunts prompt-injection inside emails.

1.3 Steps that always need a human

1. **Approving any purchase order and its size** — Almonds: formula says 150 kg, yet lead time (10 d) exceeds cover (6 d), so size alone cannot fix it.
2. **Alternate / spot suppliers, premium freight** — a cost vs stock-out trade-off.
3. **Releasing every payment batch** — maker–checker; automation only recommends.
4. **Every exception** (rate, over-billing, part-pay, credit note) — a negotiation, not arithmetic.
5. **Changing a vendor bank account** — highest-fraud step; call-back to the number on file, two approvers, cooling-off.
6. **Duplicate overrides and sending external email** — the app creates drafts only.
7. **Anything the extractor is unsure about** — failed validation goes to an exception queue.

2 · Issue 1 — Raw Material Reorder Alerts

2.1 How the build works

STEP	WHO	WHAT
1 Extract	Gemini	Reads both supplier emails → JSON facts only (prompt in `prompts/`, server in `app/server/index.js`).
2 Validate	Code	Quote verbatim in source; PO exists in sheet; qty matches; independent regex cross-check agrees.
3 Calculate	Code	Cover = stock ÷ usage; ROP = usage × lead + safety; PO counts only if ETA ≤ cover; CRITICAL / REORDER / OK; qty = usage × 14 + safety − (stock + all open PO).
4 Approve	Human	Procurement head reviews alert and supplier email; the app never sends.

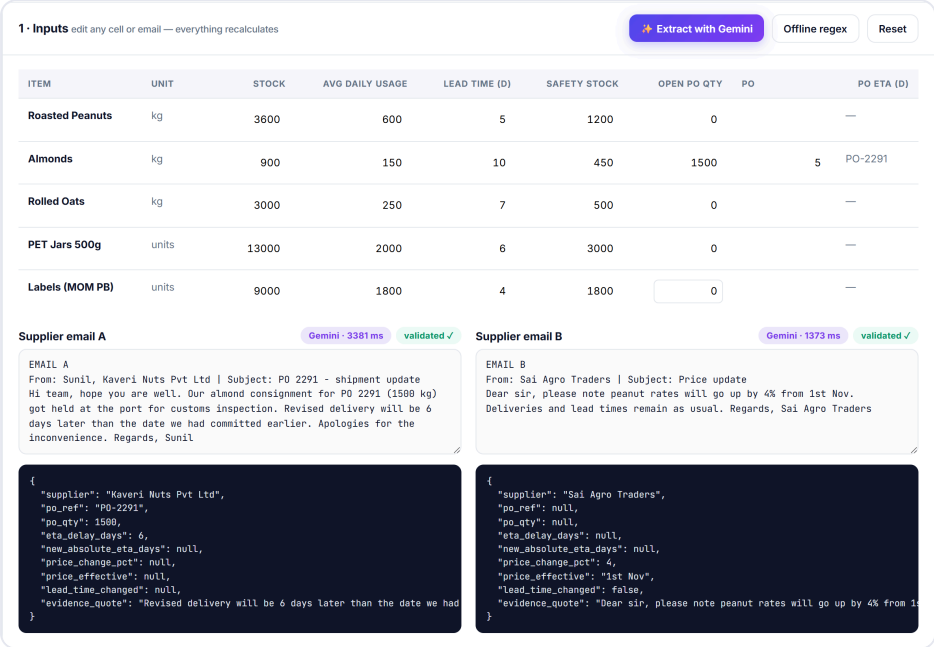


Figure 2 — Inputs: stock table, the two supplier emails and the live Gemini extraction (model latency and "validated ✓" badges shown; extracted JSON beneath each email).

Key reading of Email A: "6 days later than the date we had committed" is a *relative delay*, so the ETA becomes 5 + 6 = **11 days**, not 6. Email B (+4% peanut price from 1 Nov, lead times unchanged) changes no stock number and is raised as a price flag.

2 · What the emails changed audit log — every update is traceable

- UPDATE Almonds: PO-2291 ETA 5d → 11d (+6d, Kaveri Nuts Pvt Ltd)
- FLAG Sai Agro Traders: price +4% from 1st Nov. Lead times unchanged → no stock/ROP change; buy-ahead decision for procurement.

> **Almonds** flipped OK → CRITICAL. The PO's ETA moved from day 5 to day 11, past the 6 days of cover, so it no longer counts as available stock.

Figure 3 — Audit log of what the emails changed. Almonds flip OK → CRITICAL because the PO now lands after stock-out.

2.2 Item-level results

3 · Item-level results days of cover · ROP · status · suggested qty · action

ITEM	STOCK	DAYS OF COVER	ROP	PO COUNTED	STATUS	SUGGESTED QTY	RECOMMENDED ACTION
Roasted Peanuts	3,600	6	4,200	0	REORDER	**6,000 kg**	Place PO for 6,000 kg today; 6d cover vs 5d lead time (stock 3,600 <= ROP 4,200).
Almonds	900	6	1,950	0	CRITICAL	**150 kg**	Stock-out in 6d vs 10d lead time; PO-2291 lands day 11 (gap 5d = ~750 kg). Expedite / part-ship the PO and source spot cover now; formula qty 150 kg - human to confirm size.
Rolled Oats	3,000	12	2,250	0	OK	—	No action; 12d cover, stock 3,000 above ROP 2,250.
PET Jars 500g	13,000	6.5	15,000	0	REORDER	**18,000 units**	Place PO for 18,000 units today; 6.5d cover vs 6d lead time (stock 13,000 <= ROP 15,000).
Labels (MOM PB)	9,000	5	9,000	0	REORDER	**18,000 units**	Place PO for 18,000 units today; 5d cover vs 4d lead time (stock 9,000 <= ROP 9,000).

Judgement flag: Almonds' formula quantity is small because the delayed PO is still counted as inbound for the order formula; but it cannot land within the 6 days of cover — a human must choose expedite / part-ship / spot-buy.

Figure 4 — Days of cover, ROP, status, suggested quantity and a one-line action for every item.

ITEM	DAYS COVER	ROP	STATUS	SUGGESTED QTY
Roasted Peanuts	6	4,200	REORDER	6,000 kg
Almonds	6	1,950	**CRITICAL**	150 kg
Rolled Oats	12	2,250	OK	—
PET Jars 500g	6.5	15,000	REORDER	18,000 units
Labels (MOM PB)	5	9,000	REORDER	18,000 units

Hand check (Almonds): 900 ÷ 150 = 6 · ROP 150×10+450 = 1,950 · qty (150×14+450) − (900+1,500) = 150. The delayed PO (day 11) does not count toward cover because 11 > 6.

Judgement flags: (1) Almonds' formula quantity is small because all open PO quantity is deducted, but that PO cannot land inside 6 days of cover — a human must choose expedite / part-ship / spot-buy (~750 kg gap). (2) Labels sit exactly at ROP with only 1 day of margin over lead time. (3) Peanuts should be ordered before the +4% price rise on 1 Nov.

2.3 Procurement alert (72 words, limit 100)

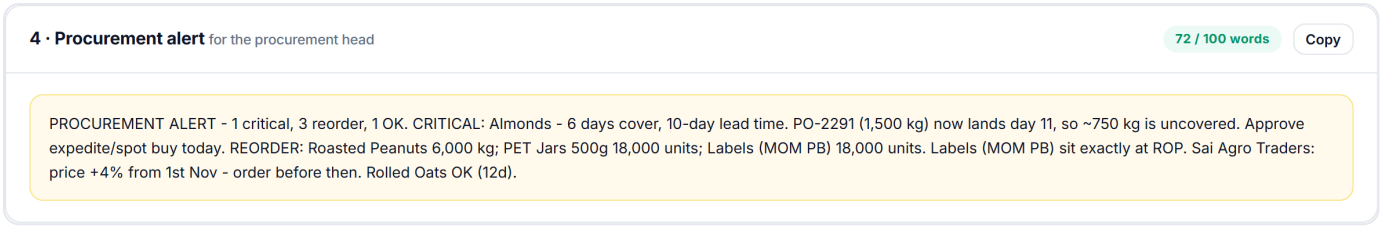


Figure 5 — Alert with automatic word-count check.

PROCUREMENT ALERT - 1 critical, 3 reorder, 1 OK. CRITICAL: Almonds - 6 days cover, 10-day lead time. PO-2291 (1,500 kg) now lands day 11, so ~750 kg is uncovered. Approve expedite/spot buy today. REORDER: Roasted Peanuts 6,000 kg; PET Jars 500g 18,000 units; Labels (MOM PB) 18,000 units. Labels (MOM PB) sit exactly at ROP. Sai Agro Traders: price +4% from 1st Nov - order before then. Rolled Oats OK (12d).

2.4 Draft email to supplier — most urgent item: Almonds (Kaveri Nuts)

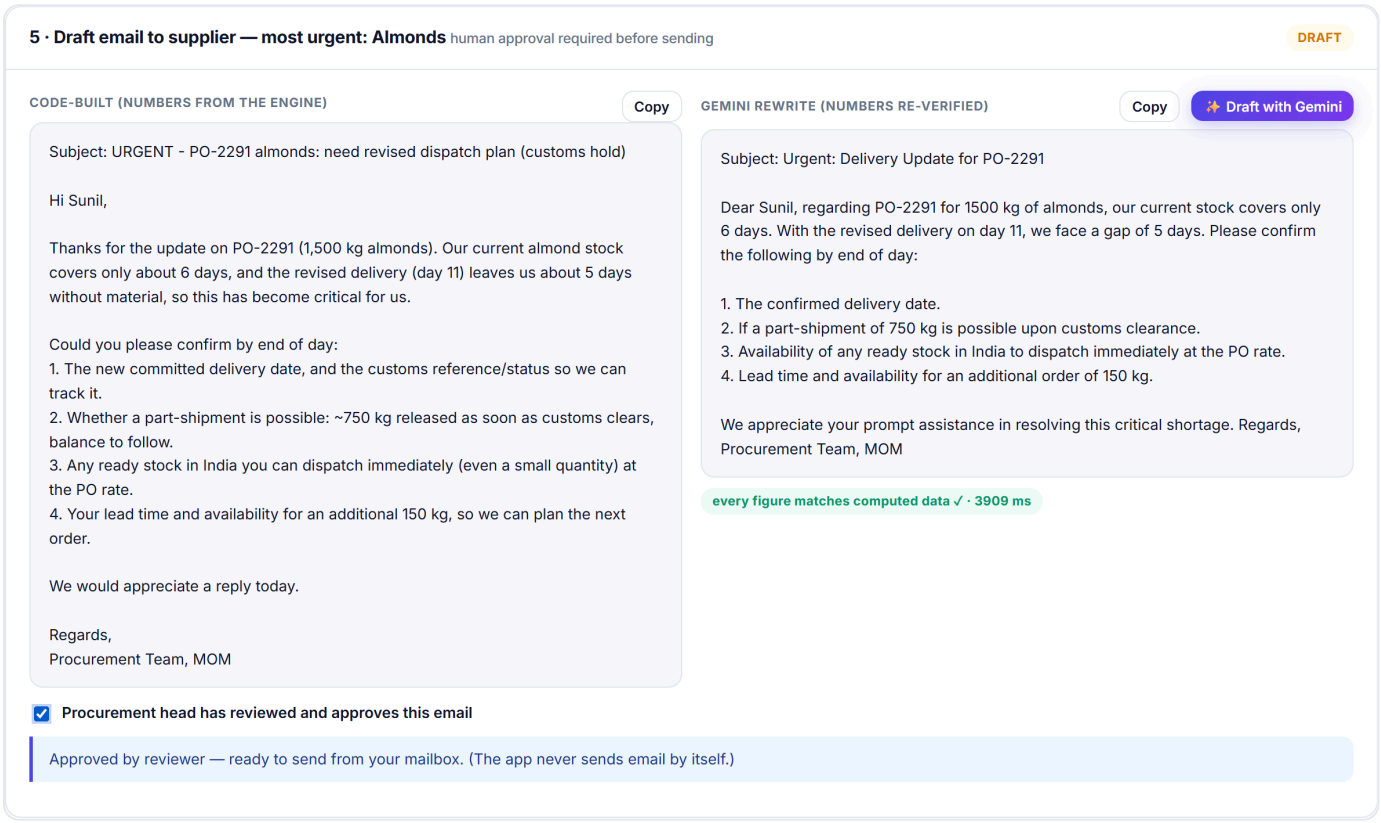


Figure 6 — Left: code-built draft (all figures from the engine). Right: live Gemini rewrite, automatically checked — “every figure matches computed data ✓”. Reviewer approval checkbox is required before sending.

3 · Issue 2 — Invoice Matching & Exception Report

3.1 How the build works

STEP	WHO	WHAT
1 Extract	Gemini	Five documents in five layouts (*24-Sep-2026*, *25/09/2026*, *26 Sep 26*; *PO-2285* vs *PO 2287*) → one JSON schema.
2 Validate	Code	qty × rate = taxable; taxable + GST = total; total present in source; independent regex cross-check on invoice no., PO, qty, rate, taxable, total.
3 Match	Code	PO exists · vendor/item agree · billed rate = PO rate · billed qty ≤ GRN · invoice number never seen before.
4 Release	Human	Finance Controller approves the payment batch; bank changes are never accepted from email.

2 · Finance inbox — 5 documents edit text to test other layouts

Doc 1 Gemini · 1775 ms validated ✓

```
DOCUMENT 1
Shree Packaging Pvt Ltd | Tax Invoice No: SP/2026/0871 | Date 24-Sep-2026
| PO Ref: PO-2285 | Item: PET Jar 500g x 20,000 nos @ Rs 11.50 | Taxable
value: Rs 230,000 | GST 18%: Rs 41,400 | Total: Rs 271,400
```

```
{
  "doc_type": "invoice",
  "vendor": "Shree Packaging Pvt Ltd",
  "invoice_no": "SP/2026/0871",
  "invoice_date": "2026-09-24",
  "po_no": "PO-2285",
  "item": "PET Jar 500g",
  "qty": 20000,
  "unit": "nos",
  "rate": 11.5,
  "taxable": 230000,
  "gst_pct": 18,
  "gst_amt": 41400,
```

Doc 2 Gemini · 3617 ms validated ✓

```
DOCUMENT 2
Om Print Labels - INV OPL-4410 dt 25/09/2026. Against PO-2286. Labels (MOM
PB) qty 30000 rate 1.35/pc. Taxable 40,500. GST @18% 7,290. Invoice total
47,790
```

```
{
  "doc_type": "invoice",
  "vendor": "Om Print Labels",
  "invoice_no": "OPL-4410",
  "invoice_date": "2026-09-25",
  "po_no": "PO-2286",
  "item": "Labels (MOM PB)",
  "qty": 30000,
  "unit": "pc",
  "rate": 1.35,
  "taxable": 40500,
  "gst_pct": 18,
  "gst_amt": 7290,
```

Doc 3 Gemini · 5068 ms validated ✓

```
DOCUMENT 3
SAI AGRO TRADERS - Bill no. SAT/889 - 26 Sep 26 - PO 2287 - Roasted
Peanuts 5000 kg x Rs 95/kg = 475,000. GST 5% = 23,750. Net payable Rs
498,750
```

```
{
  "doc_type": "invoice",
  "vendor": "Sai Agro Traders",
  "invoice_no": "SAT/889",
  "invoice_date": "2026-09-26",
  "po_no": "PO-2287",
  "item": "Roasted Peanuts",
  "qty": 5000,
  "unit": "kg",
  "rate": 95,
  "taxable": 475000,
  "gst_pct": 5,
  "gst_amt": 23750,
```

Doc 4 Gemini · 4616 ms validated ✓

```
DOCUMENT 4
Ganga Oats Mills, Invoice GOM/1207, 26-09-2026, PO-2289, Rolled Oats 3000
kg @ 58 per kg, taxable Rs 174,000, GST 5% Rs 8,700, total Rs 182,700
```

```
{
  "doc_type": "invoice",
  "vendor": "Ganga Oats Mills",
  "invoice_no": "GOM/1207",
  "invoice_date": "2026-09-26",
  "po_no": "PO-2289",
  "item": "Rolled Oats",
  "qty": 3000,
  "unit": "kg",
  "rate": 58,
  "taxable": 174000,
  "gst_pct": 5,
  "gst_amt": 8700,
```

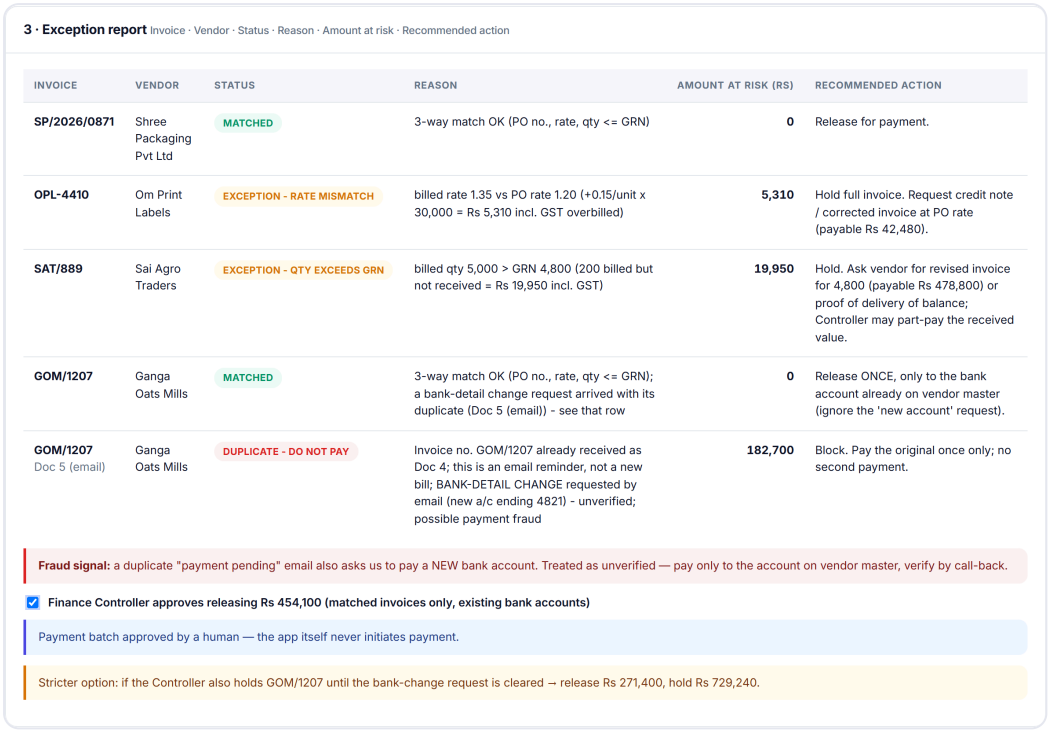
Doc 5 (email) Gemini · 1841 ms validated ✓

```
DOCUMENT 5 (email)
Subject: REMINDER - payment pending, Invoice GOM/1207
Dear Accounts, attaching invoice GOM/1207 dated 26-09-2026 for Rs 182,700
(PO-2289, Rolled Oats 3000 kg) again as payment is pending. Also please
note our bank account has changed - kindly release payment to our new
account ending 4821. Regards, Ganga Oats Mills accounts
```

```
{
  "doc_type": "email",
  "vendor": "Ganga Oats Mills",
  "invoice_no": "GOM/1207",
  "invoice_date": "2026-09-26",
  "po_no": "PO-2289",
  "item": "Rolled Oats",
  "qty": 3000,
  "unit": "kg",
  "rate": null,
  "taxable": null,
  "gst_pct": null,
  "gst_amt": null,
```

Figure 7 — All five finance-inbox documents with live Gemini extraction and validation badges.

3.2 Exception report



3 · Exception report Invoice · Vendor · Status · Reason · Amount at risk · Recommended action

INVOICE	VENDOR	STATUS	REASON	AMOUNT AT RISK (RS)	RECOMMENDED ACTION
SP/2026/0871	Shree Packaging Pvt Ltd	MATCHED	3-way match OK (PO no., rate, qty <= GRN)	0	Release for payment.
OPL-4410	Om Print Labels	EXCEPTION - RATE MISMATCH	billed rate 1.35 vs PO rate 1.20 (+0.15/unit x 30,000 = Rs 5,310 incl. GST overbilled)	5,310	Hold full invoice. Request credit note / corrected invoice at PO rate (payable Rs 42,480).
SAT/889	Sai Agro Traders	EXCEPTION - QTY EXCEEDS GRN	billed qty 5,000 > GRN 4,800 (200 billed but not received = Rs 19,950 incl. GST)	19,950	Hold. Ask vendor for revised invoice for 4,800 (payable Rs 478,800) or proof of delivery of balance; Controller may part-pay the received value.
GOM/1207	Ganga Oats Mills	MATCHED	3-way match OK (PO no., rate, qty <= GRN); a bank-detail change request arrived with its duplicate (Doc 5 (email)) - see that row	0	Release ONCE, only to the bank account already on vendor master (ignore the 'new account' request).
GOM/1207 Doc 5 (email)	Ganga Oats Mills	DUPLICATE - DO NOT PAY	Invoice no. GOM/1207 already received as Doc 4; this is an email reminder, not a new bill; BANK-DETAIL CHANGE requested by email (new a/c ending 4821) - unverified; possible payment fraud	182,700	Block. Pay the original once only; no second payment.

Fraud signal: a duplicate "payment pending" email also asks us to pay a NEW bank account. Treated as unverified — pay only to the account on vendor master, verify by call-back.

☑ **Finance Controller approves releasing Rs 454,100 (matched invoices only, existing bank accounts)**

Payment batch approved by a human — the app itself never initiates payment.

Stricter option: if the Controller also holds GOM/1207 until the bank-change request is cleared → release Rs 271,400, hold Rs 729,240.

Figure 8 — Exception report with the six required columns: Invoice · Vendor · Status · Reason · Amount at risk (Rs) · Recommended action.

INVOICE	FINDING	AT RISK (RS)
SP/2026/0871	Matched (rate 11.50, qty 20,000 = GRN).	0
OPL-4410	Rate 1.35 vs PO 1.20 → +4,500 + 18% GST.	5,310
SAT/889	Billed 5,000 kg vs GRN 4,800 → 200 kg × 95 + 5% GST.	19,950
GOM/1207	Matched — release once, to bank account on vendor master only.	0
GOM/1207 (Doc 5)	**Duplicate** reminder + request to pay a **new bank account ending 4821** — suspected payment fraud.	182,700

3.3 Totals



TOTAL SUBMITTED
Rs 1,000,640
4 unique invoices · Rs 1,183,340 across 5 documents

SAFE TO RELEASE NOW
Rs 454,100
matched · bank a/c on master only

ON HOLD
Rs 546,540
exceptions awaiting vendor

DUPLICATE BLOCKED
Rs 182,700
never payable twice

Figure 9 — Totals computed in code (reconciled: safe + hold = total submitted).

Total submitted Rs 1,000,640 (4 unique invoices; Rs 1,183,340 across 5 documents) · **Safe to release now** Rs 454,100 (271,400 + 182,700) · **On hold** Rs 546,540 (47,790 + 498,750) · **Duplicate blocked** Rs 182,700. Stricter option if GOM/1207 is also held until the bank-change is verified: release Rs 271,400, hold Rs 729,240.

3.4 Draft 1 — query to vendor (Sai Agro Traders, SAT/889)

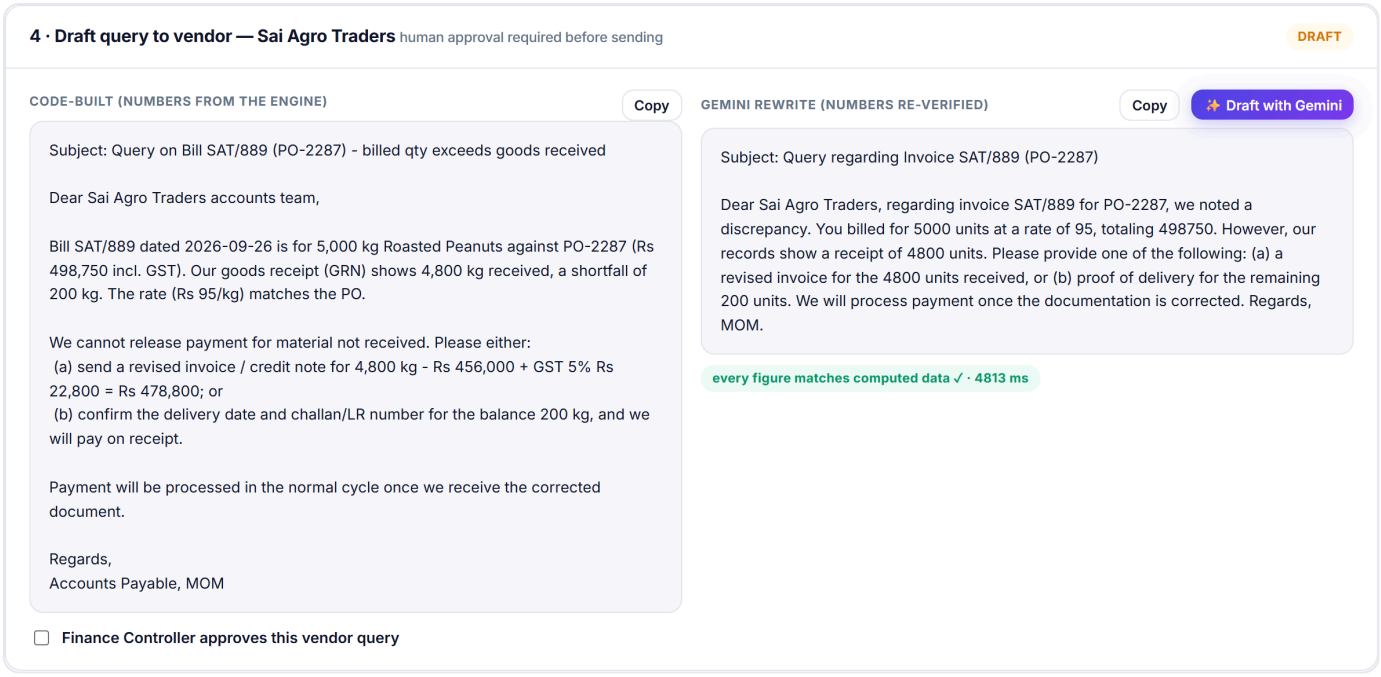


Figure 10 — Vendor query: code-built draft and Gemini rewrite with figure verification.

3.5 Draft 2 — internal note to the Finance Controller

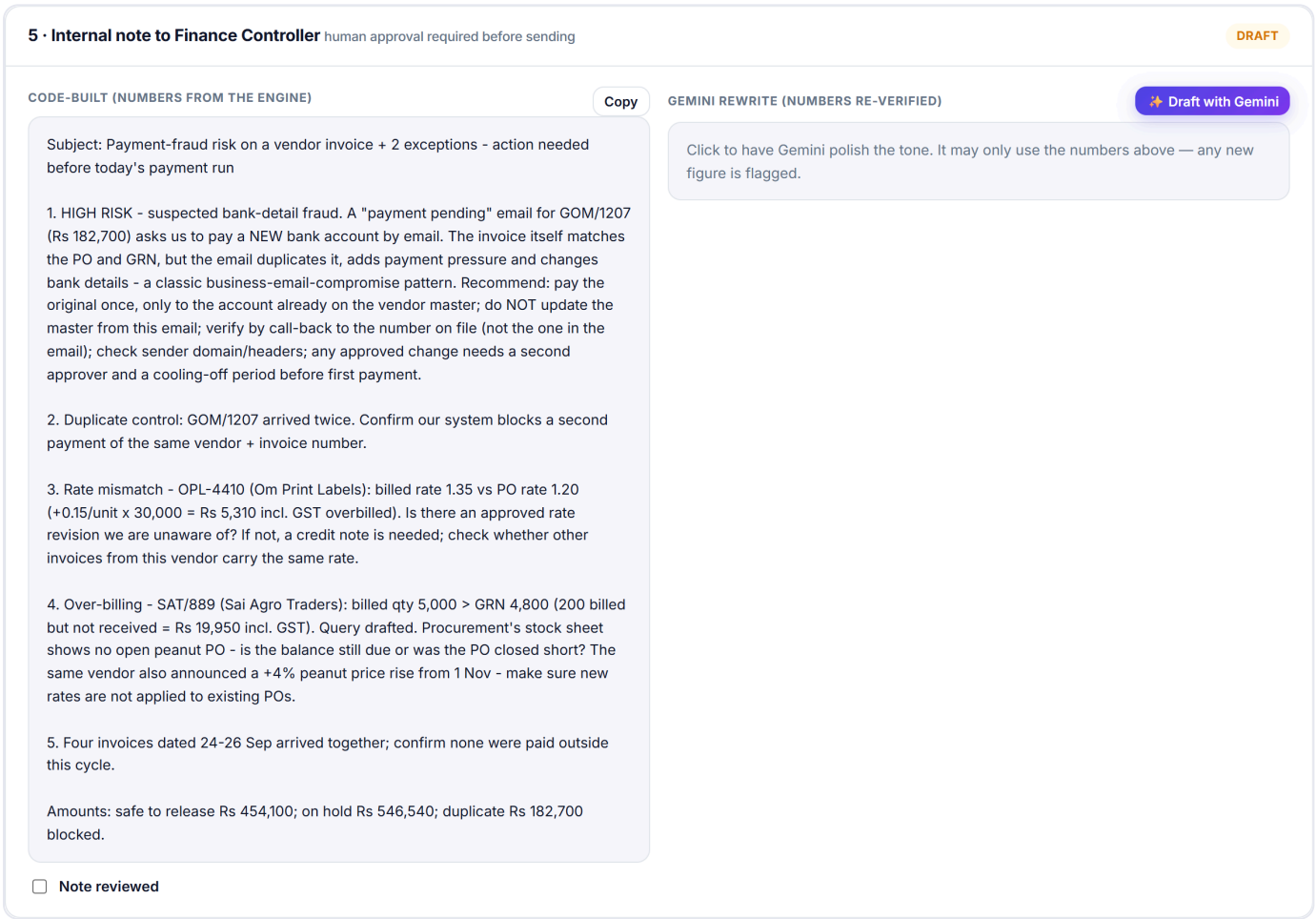


Figure 11 — Concerns beyond simple matching: bank-detail fraud pattern, duplicate control, rate and quantity exceptions, open-PO data question, peanut price rise.

4 · Part 4 — Reliability note

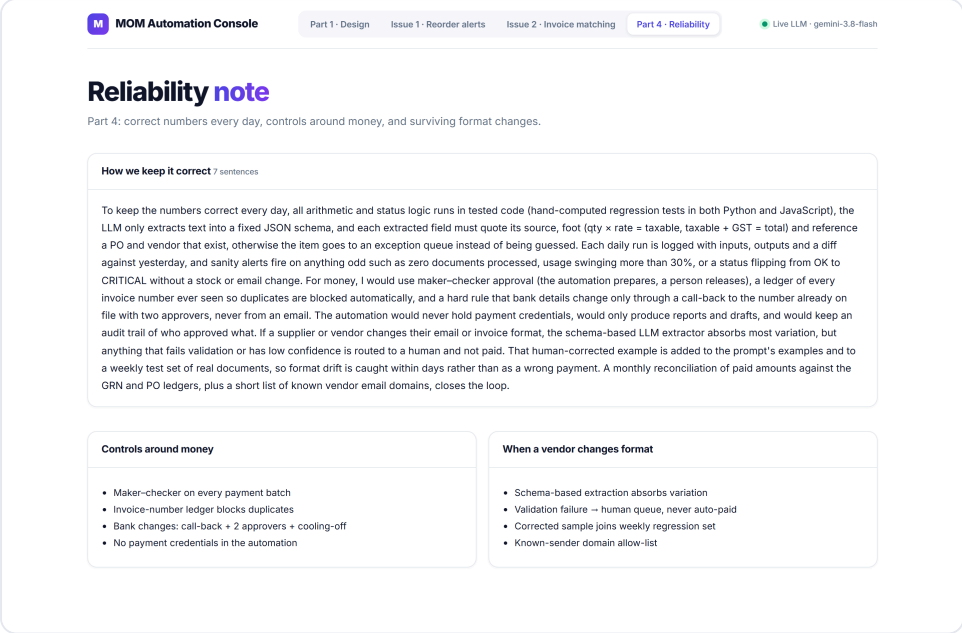


Figure 12 — Reliability tab of the console.

To keep the numbers correct every day, all arithmetic and status logic runs in tested code (hand-computed regression tests in both Python and JavaScript), the LLM only extracts text into a fixed JSON schema, and each extracted field must quote its source, foot (qty × rate = taxable, taxable + GST = total) and reference a PO and vendor that exist, otherwise the item goes to an exception queue instead of being guessed. Each daily run is logged with inputs, outputs and a diff against yesterday, and sanity alerts fire on anything odd such as zero documents processed, usage swinging more than 30%, or a status flipping from OK to CRITICAL without a stock or email change. For money, I would use maker–checker approval (the automation prepares, a person releases), a ledger of every invoice number ever seen so duplicates are blocked automatically, and a hard rule that bank details change only through a call-back to the number already on file with two approvers, never from an email. The automation would never hold payment credentials, would only produce reports and drafts, and would keep an audit trail of who approved what. If a supplier or vendor changes their email or invoice format, the schema-based LLM extractor absorbs most variation, but anything that fails validation or has low confidence is routed to a human and not paid. That human-corrected example is added to the prompt's examples and to a weekly test set of real documents, so format drift is caught within days rather than as a wrong payment. A monthly reconciliation of paid amounts against the GRN and PO ledgers, plus a short list of known vendor email domains, closes the loop.

5 · Validation, reproduction and repository

5.1 Reliability controls built into the app

- **Three guard-rails on every LLM output:** verbatim-quote check, business validation (PO exists, qty matches, invoice foots) and cross-check against an independent offline extractor — disagreement blocks the extraction.
- **Bug caught during live testing:** Gemini once wrapped Email A's JSON in an array, which silently left Almonds "OK". The server now unwraps it and the validator rejects non-objects — exactly the failure these controls exist to catch.
- **Free-tier resilience:** model fallback chain with retry, then offline regex fallback, so the workflow never dead-ends.
- **Number-verified drafts:** any figure in a Gemini draft that is not in the computed data is flagged and must not be used.
- **Humans in charge:** approval checkboxes for each email and for the payment batch; the app never sends email or initiates payment.

5.2 Live end-to-end test result

All 2 emails and 5 invoices were extracted by Gemini and passed through the engine; results equal the independent Python implementation and the hand calculations: Almonds CRITICAL, Peanuts / PET Jars / Labels REORDER (6,000 / 18,000 / 18,000), Rolled Oats OK; invoice statuses MATCHED / RATE MISMATCH / QTY EXCEEDS GRN / MATCHED / DUPLICATE; Rs 454,100 safe, Rs 546,540 hold, Rs 182,700 blocked.

5.3 Run it

```
cd app
npm install
# create ../.env from .env.example  (GEMINI_API_KEY from aistudio.google.com)
npm run dev      # console at http://localhost:5173
npm test         # JS engine vs hand-computed numbers
python run_all.py && python -m unittest discover tests   # Python reference + tests
```

5.4 Repository layout

`app/`	React + Vite console, Node Gemini gateway, JS rules engine and tests, screenshot script
`src/` · `tests/` · `run_all.py`	Python reference implementation of both automations and hand-checked tests
`prompts/`	LLM extraction prompts, schemas and guard-rails
`data/` · `outputs/`	Inputs exactly as in the PDF; generated reports, CSVs and drafts
`docs/`	This document and all screenshots (`docs/screenshots/`)

Repository: github.com/aniket162002/mom_assignment